VOLTARAX

VoltVision AI — AI-Integrated BioSolar Panel Platform

Project Reference Document

Status, roadmap, team, IP, and the commercialization & fundraising plan

Prepared for	Internal reference — Voltarax founding team
Stage	Pre-incubated at DTU Innovation & Incubation Foundation (DTU-IIF)
Home institution	Delhi Technological University (DTU)
Pilot site	Jaipur, Rajasthan (32-panel installation)
Document purpose	Single reference covering what is built, what remains, and the exact path to formal incorporation, IP completion, and first external funding

This document is for internal and incubator/mentor use. It intentionally stays at the product, business, and technology-capability level — implementation detail is left out by design.

1. Executive Summary

Voltarax is a deeptech startup building **VoltVision AI**, a monitoring and optimisation platform for bio-coated solar panels. The core innovation is a bio-fluorescent coating, derived from fruit-peel waste (a mango and pomegranate extract blend), that converts otherwise-wasted ultraviolet light into visible light a solar cell can use. VoltVision AI is the software and sensor layer that watches that coating, the panel, and the energy it produces, and gives a site operator one dashboard to see all of it.

The company is presently in the **pre-incubation** stage at the DTU Innovation & Incubation Foundation (DTU-IIF), a Section-8 Technology Business Incubator of Delhi Technological University. One patent application has been filed; two further claims are drafted and awaiting filing. A 32-panel pilot site is designed for Jaipur, Rajasthan, and the software platform — dashboard, data pipeline, and an automated AI engine — is built and tested ahead of that hardware going live.

This document has three jobs: record exactly what is complete today, record exactly what remains before a funded, multi-site commercial rollout, and lay out — in order — the specific incubation, IP, government scheme, and fundraising steps that follow from being pre-incubated at DTU-IIF today.

2. Project Overview

2.1 The problem

A conventional silicon solar panel converts roughly 15–22% of the sunlight that reaches it into electricity. Ultraviolet light — about 8–9% of the solar spectrum — is invisible to the panel and is simply wasted. Separately, India's municipal waste stream is roughly half organic matter, most of it with no productive second use.

2.2 The solution

Voltarax's coating is a fruit-peel-waste extract (60% mango, 40% pomegranate) that absorbs UV light and re-emits it as visible light the panel can convert. Internal validation on coated samples recorded approximately **11.5% higher output under direct sunlight** and approximately **72.7% higher output under UV-dominant conditions** compared to an uncoated panel.

A coating is only useful if someone can tell when it needs attention. VoltVision AI is the monitoring layer: a sensor suite on each panel feeds a cloud platform that tracks coating condition, panel health, and energy output, and an AI engine turns that history into forecasts, fault warnings, and maintenance guidance — all surfaced on one web dashboard.

2.3 Target segments

- Institutional campuses with existing or planned solar installations (the DTU relationship sits here).
- Hospitality and large-footprint commercial sites with high rooftop solar potential and a strong interest in sustainability reporting (the ITC Hotels engagement sits here).
- Remote, off-grid, and disaster-relief deployments, where a coating that needs fewer manual cleaning visits and a monitoring system that flags faults early are worth the most.

3. Current Status — What Has Been Completed

3.1 Research & coating

- Coating formulation fixed: 60% mango peel extract / 40% pomegranate peel extract.
- Lab-scale validation complete, with the sunlight and UV output figures above recorded.
- Application and sensing method defined: the coating's condition is read via multi-spectral sensing (covering ultraviolet through near-infrared) so degradation can be scored, not just guessed at.

3.2 Hardware design

- Full sensor specification complete for a per-panel monitoring node: electrical output (voltage/current/power), ambient and panel-surface temperature, humidity, light intensity, and airborne dust.
- A camera-based inspection module is specified for visual condition checks (dust, physical damage, coating state) alongside the electrical sensors.
- A documented, step-by-step hardware build-and-wiring guide exists, including known circuit pitfalls and their fixes, ready to hand to whoever assembles the physical units.
- Hardware has **not yet been physically deployed** — see Section 4.

3.3 Software platform

- A structured data contract is defined for every sensor reading, so hardware, the cloud database, and the AI engine all agree on one format with no ambiguity.
- A cloud-based monitoring backend is built: panel status, live readings, historical trends, maintenance alerts, and carbon/sustainability reporting.
- A role-based web dashboard is built (administrator, operator, and read-only viewer access), ready to show live data the moment hardware is connected.
- An automated AI engine is built and tested, covering three capabilities: short-horizon energy forecasting, unsupervised fault/anomaly detection, and an adaptive panel-health baseline. It runs on a schedule, trains itself once enough real data exists, and needs no manual step when that happens — see Section 4 and Section 5 for exactly what "not enough data yet" means today.
- A corporate website is live, covering the product, the technology at a conceptual level, sustainability messaging, team, and a request-a-demo contact flow.

3.4 Team, IP, and institutional standing

- A six-person team is in place, spanning leadership, technology, and product execution, plus one formal mentor/advisor (Section 6 has the full list).
- One patent application has been filed; two further claims are drafted and not yet filed (Section 7).
- The project is pre-incubated at DTU-IIF.
- Delhi Technological University is the home institution; ITC Hotels is an active institutional engagement (Section 8).

4. What's Left To Do

Ordered roughly as it should happen — later items generally depend on earlier ones.

Item	What it involves	Depends on
Physical hardware deployment	Assemble and install the sensor nodes at the pilot site; this is the single gating item for everything downstream, since the AI engine and live dashboard both need real readings to work with.	—
Real-data validation window	A sustained period (recommended: 60–90 days) of live readings, used to confirm sensor accuracy against a multimeter/known reference and to give the AI engine enough history to train on.	Hardware deployment
AI models move from "waiting for data" to trained	Already automatic once enough real history exists (Section 3.3) — this is a waiting period, not a build task.	Real-data validation window
Multi-site monitoring	Today the platform is built for one site. Monitoring separate installations (e.g. Mumbai, Delhi, Haryana, Kashmir) from one dashboard needs: (a) each site's data kept separate in storage, (b) every dashboard query scoped to one site, and (c) a site switcher in the dashboard itself, plus an optional combined fleet-level view. See Section 5.	Hardware deployment (at each new site)
Complete the two pending patent claims	Draft claims exist; they need to be formally filed before any further public disclosure (investor conversations under NDA are fine; public pitch decks, demo days, or press are not, until filed). See Section 7 and Section 9.	—, time-sensitive
Formal company incorporation	Register as a Private Limited Company — required before DPIIT recognition and most funding schemes can be applied for. See Section 9.	—
DPIIT Startup India recognition	Unlocks patent-fee rebates, the Seed Fund Scheme, and most other startup-specific government support. See Section 9.	Incorporation
Progress from pre-incubation to full incubation at DTU-IIF	DTU-IIF runs a two-stage programme; moving to the incubation stage typically brings more formal mentorship, funding access, and space.	Hardware/validation progress to show
First paying pilot / revenue validation	Convert an institutional relationship (DTU campus, or a hospitality site such as ITC) into a paid or co-funded pilot, to evidence commercial demand to investors.	Hardware deployment, validation data

5. Monitoring Multiple Sites (Mumbai, Delhi, Haryana, Kashmir)

This section answers directly: if panels are installed in Mumbai, Delhi, Haryana, and Kashmir, can they be monitored from the current software?

Not cleanly, not yet. Each sensor reading already carries a site identifier in its data, but the platform's storage and dashboard were both built assuming a single installation. If four sites started sending data

today, all four would land in one shared pool with no dashboard view to tell them apart — useful for nothing more than "something, somewhere, is producing power."

5.1 What's needed for real multi-site monitoring

- **Separate storage per site** — each site's live readings, history, alerts, and AI predictions kept in their own space rather than one shared pool.
- **Site-aware backend** — every status, forecast, and alert request scoped to a specific site rather than assuming there is only one.
- **A site switcher in the dashboard** — choose Mumbai, Delhi, Haryana, or Kashmir to see that installation specifically, plus an optional combined "fleet" view across all sites at once (useful for exactly this four-city scenario, and for an investor-facing view of total deployed capacity).

This is a scoped, well-understood piece of work, not a rebuild — it should be sequenced right after the first site's hardware deployment and before a second site goes live, so it's in place by the time a third and fourth site are added.

6. Team

Name	Role	Focus
Shubhika	Founder & Managing Director	Company direction and renewable-energy innovation
Dinesh Kumar	Team Member	AI, software, and cybersecurity
Pradeep Patel	Team Member	Engineering and product development
Rohan	Team Member	Product and execution
Manodeep	Team Member	Product and execution
Dr. Jai Gopal Sharma	Mentor & Startup Advisor	Mentorship and startup guidance

A full founder/team photo-and-bio set for investor materials is a Section 9 action item, not yet assembled.

7. Intellectual Property

Claim	Covers	Status
Claim 1	Spectral monitoring of bio-coating condition, to predict degradation and recommend recoating timing.	**Filed**
Claim 2	Combined visual inspection method covering both UV-related coating condition and ordinary physical surface defects.	**Drafted — not filed**
Claim 3	Adaptive energy-management/dispatch method that uses coating health as an input, not just weather and battery charge.	**Drafted — not filed**

Standing caution: Claims 2 and 3 must be formally filed before they are described in any public setting — a pitch to a closed, NDA-bound investor audience is generally fine; a public demo day, open pitch competition, published deck, or press mention is not, until filing is complete. Section 9 gives the exact filing route and its cost/timeline.

8. Institutional Partners & Clients

Name	Relationship
Delhi Technological University (DTU)	Home institution; site of ongoing research and the project's institutional base
ITC Hotels	Institutional/hospitality client engagement
Others	Additional institutional and hospitality partnerships in active discussion — not yet finalised for public listing

9. Commercialization & Fundraising Roadmap

This section is written specifically for where the project stands today: **pre-incubated at DTU-IIF**, one patent filed, hardware not yet deployed. Each step below names the exact scheme or body it concerns, its current eligibility rule, and what it unlocks. It is ordered — later steps generally need earlier ones done first.

Step 1 — Confirm and progress DTU-IIF pre-incubation status

DTU-IIF runs incubation in two formal stages: **pre-incubation** (where this project currently sits) and **incubation** proper. Ask the DTU-IIF team directly: (a) what evidence is needed to move from pre-incubation to full incubation, and (b) whether DTU-IIF is registered as a **NIDHI-PRAYAS Centre** under the Department of Science & Technology — DTU-IIF has run DST/DST-NIDHI-backed programmes before, so this is worth asking about directly rather than assuming either way.

Step 2 — Incorporate the company formally

Register as a Private Limited Company (the standard structure for a startup taking on investment in India). This is a hard prerequisite for Step 3 onward — DPIIT recognition, patent fee rebates, and every funding scheme below all require a registered legal entity, not an informal team.

Step 3 — Obtain DPIIT Startup India recognition

Apply via the Startup India portal once incorporated. This is normally processed in about 5–7 working days and is a prerequisite for every benefit in Steps 4 and 5. There is no cost to apply.

Step 4 — Complete the patent filings via the SIPP scheme

The Scheme for Facilitating Startups' Intellectual Property Protection (SIPP) gives a DPIIT-recognised startup a government-empanelled patent facilitator at **zero professional fee** — the facilitator's fee is paid by the government, not by the startup — plus an **80% rebate** on the statutory government filing fee itself. Use this route to file Claims 2 and 3 (Section 7). A provisional application can be filed first to lock in a priority date cheaply, followed by a complete specification within 12 months.

Step 5 — Apply for early, non-dilutive grant funding

Scheme	Who runs it	What it gives	Fit today
NIDHI-PRAYAS	Dept. of Science & Technology (DST), via a registered Pre-incubation/PRAYAS Centre	Grant of roughly Rs. 10–20 lakh for prototype/proof-of-concept development	Strong fit — matches the current pre-incubation stage exactly; apply through DTU-IIF if it holds PRAYAS-Centre status
Startup India Seed Fund Scheme (SISFS)	DPIIT, disbursed through approved incubators	Up to Rs. 20 lakh grant for proof-of-concept, plus up to Rs. 50 lakh as a convertible debenture for market entry	Apply once DPIIT-recognised and incorporated less than 2 years; check which incubators near you are SISFS-approved disbursing partners

Scheme	Who runs it	What it gives	Fit today
State-level cleantech/patent incentives	State startup cells (varies by state)	Several states reimburse patent costs and offer cleantech-specific grants, on top of the central schemes above	Check the home state's startup cell (where the company is incorporated) for a current scheme list

These are non-dilutive (no equity given up) and are normally sequenced before any investor conversation — they both fund the hardware-deployment and validation work in Section 4, and make the company a more credible, de-risked pitch once Step 6 begins.

Step 6 — Deploy hardware and generate real validation data

Run the pilot deployment and the 60–90 day validation window from Section 4. This produces the single most valuable thing for an investor conversation: real performance numbers from a real site, not projections.

Step 7 — Build the investor-facing material set

- Pitch deck: problem, solution, traction (DTU, ITC Hotels, pilot data), team, IP position, market size, ask.
- Financial model: unit economics per panel/site, cost of hardware + coating per installation, projected revenue model (hardware sale, subscription/SaaS monitoring fee, or a hybrid), and a funding-use breakdown.
- Data room: cap table, incorporation documents, DPIIT certificate, patent filing receipts/status, any signed pilot/LOI documents with institutional partners.
- A short, pilot-data-backed case study or demo video from the first deployed site.

Step 8 — Approach angel and early-stage climate-tech investors

With DPIIT recognition, a filed IP position, and real pilot data in hand, approach: climate-tech and deep-tech-focused angel networks, university-affiliated or incubator-affiliated angel funds (DTU-IIF's own network is the first port of call), and early-stage VCs with a stated cleantech/climate thesis. A priced equity round or a convertible instrument (CCD/SAFE-equivalent note) are the two usual structures at this stage — which one depends on what the lead investor prefers at the time.

Ongoing, in parallel

- Udyam (MSME) registration — inexpensive, opens up government tendering and some additional state benefits.
- Basic statutory compliance: GST registration once revenue starts, and a proper bookkeeping/accounting setup from incorporation onward — investors will ask for clean books in diligence.

Appendix A. Glossary

Term	Meaning
DPIIT	Department for Promotion of Industry and Internal Trade — the central government body that grants official "Startup India" recognition.
TBI	Technology Business Incubator — the formal category DTU-IIF belongs to.
NIDHI	National Initiative for Developing and Harnessing Innovations — an umbrella set of DST startup-support schemes, of which PRAYAS is one part.
SISFS	Startup India Seed Fund Scheme — the central seed-grant scheme described in Step 5.
SIPP	Scheme for Facilitating Startups' Intellectual Property Protection — the free patent-filing-assistance scheme described in Step 4.

Appendix B. Site Rollout Checklist (per new site)

A repeatable checklist for each new installation, including the four cities under consideration.

- Site survey: roof/ground space, orientation, shading, local grid or battery setup.
- Hardware procurement and assembly for that site's panel count.
- On-site installation and sensor wiring, following the standard build guide.
- Confirm the site is correctly tagged in the data (its own identifier, kept separate from every other site — see Section 5).
- 7–14 day burn-in period: confirm every sensor reads correctly against a known reference before trusting the data.
- Add the site to the dashboard's site switcher once Section 5's multi-site work is in place.

End of document. This is a living reference — update the status sections as hardware deployment, patent filings, and funding applications progress.